

WINTERLAND, NL, Canada

WMO: 711810

Lat: 47.1365N

Lon: 55.3265W

Elev: 29

StdP: 100.97

Time Zone: -3.50 (NAF)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.5	-11.0	-17.4	0.8	-11.4	-15.6	1.0	-10.2	20.3	-2.0	18.4	-3.3	5.8	300	0.775

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.5	23.5	18.7	22.3	18.0	21.0	17.5	20.2	21.9	19.3	20.9	18.4	20.1	5.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.3	21.2	18.6	13.5	20.2	17.7	12.8	19.4	58.3	21.9	55.2	21.0	52.4	20.0	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 16.2	14.2	12.6	DB	-15.8	26.3	2.8	1.4	-17.8	27.3	-19.5	28.1	-21.0	28.9	-23.1	29.9	
(5)			WB	-16.8	22.0	2.7	0.6	-18.7	22.4	-20.3	22.8	-21.8	23.1	-23.7	23.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-3.2	-3.8	-1.6	2.8	7.4	11.5	16.3	17.1	13.4	8.8	4.3	0.2
	DBStd	7.84	3.89	4.10	3.58	2.82	2.66	2.98	2.37	2.24	3.20	3.38	3.61	3.95
	HDD10.0	2028	409	387	360	215	89	17	0	0	7	65	174	304
	HDD18.3	4466	668	621	618	465	339	204	71	50	149	296	422	562
	CDD10.0	625	0	0	0	0	8	63	195	220	109	27	3	0
	CDD18.3	22	0	0	0	0	0	1	7	12	2	0	0	0
	CDH23.3	40	0	0	0	0	0	1	18	19	2	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	5.8	6.8	7.2	6.6	6.1	5.5	5.1	4.5	4.5	5.1	5.8	5.8	6.6
Precipitation	PrecAvg	1493	129	120	112	115	119	93	103	95	136	162	156	148
	PrecMax	1750	201	184	231	289	192	164	181	136	223	267	230	203
	PrecMin	1302	88	64	64	47	41	36	55	60	45	86	77	96
	PrecStd	121	28	33	43	63	40	37	31	20	52	49	40	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	7.6	9.5	13.1	18.9	22.7	25.3	25.2	23.2	18.7	14.8	12.2
		MCWB	10.1	7.3	8.0	9.0	13.6	17.2	20.2	19.7	18.8	17.1	13.9	11.8
	2%	DB	6.7	5.4	6.7	10.9	16.4	20.6	23.4	23.7	21.2	17.3	13.1	9.9
		MCWB	6.5	5.1	5.3	8.4	11.6	16.0	18.6	19.1	18.1	16.0	12.4	9.4
	5%	DB	3.8	3.3	4.8	9.2	14.3	18.9	22.1	22.5	19.8	15.8	11.9	7.8
		MCWB	3.2	2.7	3.5	7.1	10.3	14.4	17.7	18.4	17.4	14.2	11.2	7.4
	10%	DB	2.1	1.7	3.3	7.4	12.5	17.0	20.8	21.2	18.5	14.2	10.3	6.0
		MCWB	1.4	1.0	2.0	5.5	9.4	13.4	17.1	17.9	16.3	12.7	9.6	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	7.4	8.2	11.0	15.1	19.0	21.3	21.6	20.5	17.7	14.1	12.1
		MCDB	10.1	7.5	9.0	12.2	17.2	21.5	24.1	23.4	21.7	18.3	14.6	12.4
	2%	WB	6.4	4.9	5.6	9.1	12.8	16.9	19.9	20.4	19.1	16.3	12.6	9.5
		MCDB	6.5	5.0	6.5	10.2	14.5	19.1	22.0	22.1	20.4	16.9	13.0	9.9
	5%	WB	3.4	2.6	3.7	7.5	11.2	15.5	18.8	19.5	17.9	14.5	11.4	7.4
		MCDB	3.7	3.0	4.4	8.8	13.3	17.9	20.7	21.1	19.3	15.5	11.8	7.8
	10%	WB	1.5	0.9	2.1	5.7	9.9	14.2	17.9	18.7	16.8	13.0	9.6	5.4
		MCDB	2.0	1.4	3.0	6.9	12.3	16.5	19.6	20.4	18.1	14.0	10.2	5.9
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.0	6.3	7.2	8.6	8.5	8.1	7.5	7.9	6.8	6.4	5.6
		MCDBR	8.5	7.9	8.0	9.6	11.9	11.1	10.1	9.1	8.4	7.9	8.2	8.0
		MCWBR	9.0	8.4	7.1	7.3	7.8	7.0	5.9	5.4	5.7	6.6	8.2	8.3
	5% WB	MCDBR	8.4	7.9	7.9	9.0	10.1	9.7	8.1	7.5	7.6	7.6	8.1	8.1
		MCWBR	9.0	8.3	7.6	7.7	7.6	6.9	5.5	5.4	6.0	7.1	8.3	8.4
Clear-Sky Solar Irradiance	taub	0.321	0.319	0.332	0.372	0.379	0.402	0.430	0.406	0.371	0.352	0.324	0.313	
	taud	2.341	2.425	2.431	2.343	2.385	2.349	2.320	2.388	2.465	2.466	2.472	2.385	
	Ebn at Noon	752	850	892	882	886	864	833	839	840	800	753	718	
	Edh at Noon	74	84	97	116	116	121	123	110	93	80	66	64	
All-Sky Solar Radiation	RadAvg	1.01	1.71	2.87	4.19	5.02	5.14	4.90	4.34	3.54	2.20	1.22	0.84	
	RadStd	0.08	0.17	0.18	0.37	0.38	0.52	0.36	0.27	0.22	0.16	0.15	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page